

Uzair Ahmad Khan

+92 313 5167976 | uzairahmadkhanmalik@gmail.com | [LinkedIn](#) | [GitHub](#)

SUMMARY

Computer Science graduate (CGPA: 3.83) from UET Taxila specializing in **Artificial Intelligence, Computer Vision, Deep Learning, NLP, Data Annotation, and Machine Learning**. Experienced in custom model training, object detection, object tracking, pose estimation, image processing, and real-time AI solutions using YOLO and OpenCV. Developed AI systems for traffic monitoring, industrial safety, healthcare monitoring, agriculture, fire detection, parking management, and automated quality inspection. **Open to AI Engineer and Computer Vision Engineer roles.**

EDUCATION

University of Engineering & Technology (UET), Taxila <i>Bachelor of Science in Computer Science</i>	Sep 2022 – Jun 2026 <i>CGPA: 3.83 / 4.00</i>
Fazaia Inter College Minhas, Kamra <i>FSc Pre-Engineering</i>	Jun 2020 – Jun 2022 <i>Percentage: 94%</i>

TECHNICAL SKILLS

AI & Computer Vision: Deep Learning, CNNs, Computer Vision, Object Detection, Object Tracking, Image Processing, Pose Estimation, Custom Model Training, YOLOv11, YOLOv12, YOLO26, YOLO11n, YOLOv11-OBb, EasyOCR
Machine Learning & NLP: Machine Learning, Natural Language Processing (NLP), TF-IDF, Naive Bayes, SVM, Logistic Regression
Data Annotation: Image Annotation, Object Detection Annotation, Custom Dataset Preparation, Data Labeling
Frameworks & Libraries: Python, TensorFlow, Keras, PyTorch, OpenCV, Ultralytics, NumPy, Pandas, Scikit-Learn
Deployment & Tools: React, Flask, Streamlit, Git, GitHub, SORT, ByteTrack, BoT-SORT, Supervision, n8n Automation, AI Agents

EXPERIENCE

Machine Learning Intern <i>Elevo</i>	Sep 2025 – Oct 2025 <i>Remote</i>
- Developed a CNN-based Traffic Sign Recognition model for autonomous driving applications. - Built machine learning models for Loan Approval Prediction and Student Score Prediction.	

ACADEMIC & COMPUTER VISION PROJECTS

NextGen Traffic: Smart Traffic Monitoring System <i>YOLOv11, SORT, EasyOCR, React</i>	FYP
- Built a real-time traffic monitoring system using YOLOv11 and SORT, with custom-trained models for rickshaws, cars, buses, trucks, and motorcycles, direction-aware counting, and traffic-flow analytics. - Integrated License Plate Recognition (EasyOCR) and helmet, wrong-way, triple-riding, and accident detection modules; built a React dashboard for live statistics and violation alerts. Project Link	
AI-Powered Bottle Cap Detection & Counting System <i>Python, YOLO, Computer Vision, OpenCV</i>	
- Developed a real-time inspection system classifying bottles as Capped/Uncapped with object tracking and virtual-line counting to count each bottle once. - Added live statistics and color-coded bounding boxes for automated quality inspection. Project Link	
AI-Based Skipping Counter using YOLO Pose Estimation <i>YOLO Pose, Ultralytics, OpenCV, NumPy</i>	
- Built a real-time skipping counter using pose estimation and ankle keypoint tracking, with dynamic counting-line positioning, movement smoothing, and re-arm logic. - Added jumps-per-minute calculation and skeleton visualization. Project Link	
Real-Time IV Drip Chamber Monitoring System <i>Python, Custom YOLO, OpenCV</i>	
- Developed a custom-trained YOLO system to detect the drip chamber, fluid drops, and tube in real time, with drop tracking, drop counting, and flow-rate calculation. - Added Normal/Slow/Fast flow classification with low-fluid and stopped-flow alerts. Project Link	
SkySight AI — Aerial Traffic Monitoring & Tracking <i>Python, YOLOv12, OpenCV, Supervision, ByteTrack</i>	
- Developed a custom YOLOv12 system for vehicle detection and tracking in high-altitude drone footage with Small/Large vehicle classification and traffic-density heatmaps. - Used ByteTrack for stable tracking and optimized real-time video processing. Project Link	
Smart PPE Tracking & Compliance System <i>Python, Custom YOLO26, Computer Vision</i>	
- Developed a custom YOLO26 system to detect workers and PPE, with real-time personnel tracking and per-worker safety equipment association. - Generated automated PPE compliance statistics for industrial safety and site monitoring. Project Link	

Smart Livestock Tracking & Counting System | *Python, YOLOv12, BoT-SORT*

- Built a real-time livestock detection and tracking system counting sheep across a virtual boundary, using BoT-SORT to handle occlusion in crowded scenes.
- Added a live statistics dashboard and Picture-in-Picture view of the last counted animal. [Project Link](#)

Real-Time Fire & Smoke Detection System | *Python, Custom YOLO26, OpenCV*

- Developed a custom YOLO26 fire/smoke detection system with temporal voting to reduce false positives from steam, fog, and dust.
- Added multi-channel alarm, email, and SMS alerts with live video inference. [Project Link](#)

AI Smart Parking System | *Python, YOLOv11-OBB, OpenCV, ByteTrack*

- Developed an AI parking system using YOLOv11-OBB for angled vehicle detection, dynamic slot mapping, and automatic zone-wise space calculation.
- Added a real-time parking statistics heatmap using ByteTrack. [Project Link](#)

Real-Time People Counting System | *YOLO11n, SORT, Python*

- Built a real-time retail analytics system to detect, track, and count people entering/exiting a store with consistent identity tracking for automated people-flow monitoring. [Project Link](#)

CERTIFICATIONS

Computer Vision: YOLO Custom Object Detection with Colab GPU

Coursera

Python for Data Science, AI & Development

IBM (Coursera)

Machine Learning with Python

IBM (Coursera)

The Complete MySQL from Scratch: Bootcamp

Udemy

ACHIEVEMENTS

- Scored **73 Percentile** in NSCT (National Skill Competency Test).
- Completed the **ACM UET Fellowship AI Track** focused on Artificial Intelligence research and collaboration.
- Maintained strong academic performance with **CGPA 3.83** in BS Computer Science.